Automated Reading of Skin Prick Test

Report
CS F376 DESIGN PROJECT
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CERTIFICATE

This is to certify that the Semester Project Report entitled, **Automated Reading of Skin Prick Test** and submitted by **Jannath Shaik ID No. 2021A7PS0132U** in complete fulfillment of the requirement CS F376 DESIGN PROJECT embodies the work done by her under my supervision.

Date: **02/06/2024**

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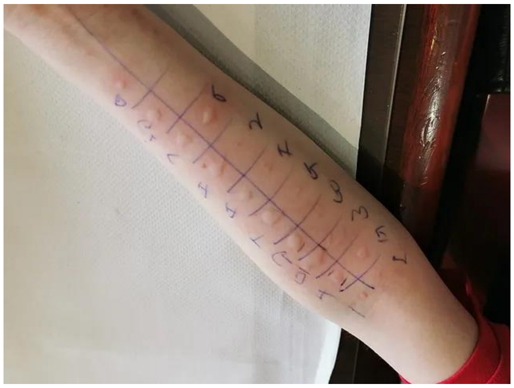
Abstract: Skin Prick Test (SPT) is an allergy test that entails the process of pricking the skin and introducing a small amount of allergen beneath the skin surface. The size of the reaction, a bump at the prick site called a wheal, indicates the severity of the allergic reaction. Nevertheless, using rulers in traditional ways to measure wheal size is subjective and can lead to mistakes because of its inconsistency among several observers and non-symmetrical wheal shapes. This paper suggests a new method that combines 3D reconstruction techniques with Mask R-CNN, a deep learning model for object detection and segmentation, in order to overcome these drawbacks. Our objective is to come up with better measurement methods that are more accurate, reliable and unbiased than the ones already existing which improve allergy diagnosis and treatment decisions.

Signature of Student: Jannath Shaik **Signature of Supervisor:**

Date: 02/06/2024 **Date: 02/06/2024**

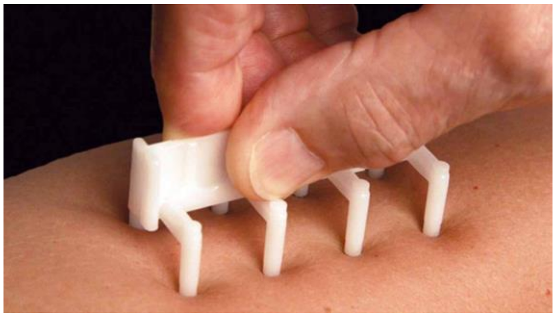
TABLE OF CONTENTS

Chapter 1: Introduction



1.1. **Wheal Formation due to Skin Prick Test**

Skin Pick Test (SPT) is commonly used as a test to establish the occurrence of any allergies mediated by immunoglobulin E (IgE). A tiny quantity of allergens is injected into the skin just below the surface, by pricking it slightly. The immune reacts to a specific allergen, if the patient is allergic to it by releasing chemicals at the prick area. The reaction takes 15-20 minutes to over, which results in a bump known as a wheal. The size of the wheal signifies the severity level of the allergen [12]. The diameter of the largest wheal is manually measured for calculating the area and diameter of the wheals.



1.2. Reading of Skin Prick Test using ruler

SPT is used in several clinical tests such as diagnosing allergic rhinitis, allergic asthma, atopic dermatitis, and food allergies. It is a cheap,simple and time efficient procedure which can test for different allergens at the same time. Additionally, medications which interfere with the blood test do not influence SPT much.

The wheal appearing on the skin may be asymmetrical in shape, which makes it difficult to manually capture the accurate dimensions of the wheals. For example, only the distance between two edges of the wheal are measured using a ruler and that distance may not be the diameter but another chord belonging to an oval wheal. This method is not precise in estimating the entire area as it does include the area covered by the curved sides of the wheal. [13], [14]

There are inconsistencies dealt with when using a ruler, as there is a difference based on how the angle at which the ruler is held. These discrepancies might arise from repeated measurements made by a single observer (intra-observer error) or from observations made by several observers (inter-observer error). This makes cross-temporal measurement comparisons difficult and unreliable. [14]

Wheal measures that are inaccurate can cause misdiagnosis and complicate treatment choices. Overestimating could result in needless limitations, while underestimating could overlook allergies. It is challenging to monitor allergy development because of unfavorable comparisons. To avoid this situation in this paper we propose a 3D estimation model, which will help in generating a 3D reconstruction of the wheal area, in order to estimate the height of these wheals. The model detects the wheal area and estimates the height of the largest wheal. Detecting the wheal and measuring its diameter is another aspect of this paper, so as to automate the reading of wheals measurements. This makes it easier to read the wheal accurately, successfully reducing the inaccuracies in defining the allergies and reducing false positive cases,.

Chapter 2: Literature Review

This paper examines current developments in image segmentation and 3D reconstruction methods to overcome these constraints and produce more precise and consistent wheal measures.

Many research works propose the use of 3D reconstruction for SPT wheal analysis. Justo et al [2] have created a laser based gadget that reconstructs the shape of an arm in 3D and detects wheal areas automatically. For this purpose, Pineda et al [3] used a wide-field imaging system with parametric surface fitting to estimate wheal diameter by employing a broad-field imaging system and introduced parametric surface fitting. Both methods proved to be more accurate than manual measurements [2], [3].

Moreover, image segmentation methods provide another potentially useful approach for wheal analysis. Parihar et al [8] compared several segmentation techniques (Otsu's, region-growing method, watershed algorithm and region watershed) using mobile phone camera images. However, their findings indicate that automatic segmentation is better in estimating the size of wheals in comparison with manual ones [8].

Recent studies are exploring deep learning techniques for wheal detection and segmentation. Peña et al [7], extended on the work by Pineda et al [3] on 3D reconstruction method and U-net style Convolutional Neural Network (CNN) for Wheal Segmentation was incorporated. This procedure managed to yield some promising results despite small data set limitations [7].

In addition to this, a 3-D model can be realized by combining volumetric and surfacic techniques. This approach is often used in commercial software, but some open-source tools also exist for the purpose of carrying out the operation [1]. To demonstrate its potentiality on alternative 3D imaging approaches , this technology has not been applied directly to SPT wheal measurement.

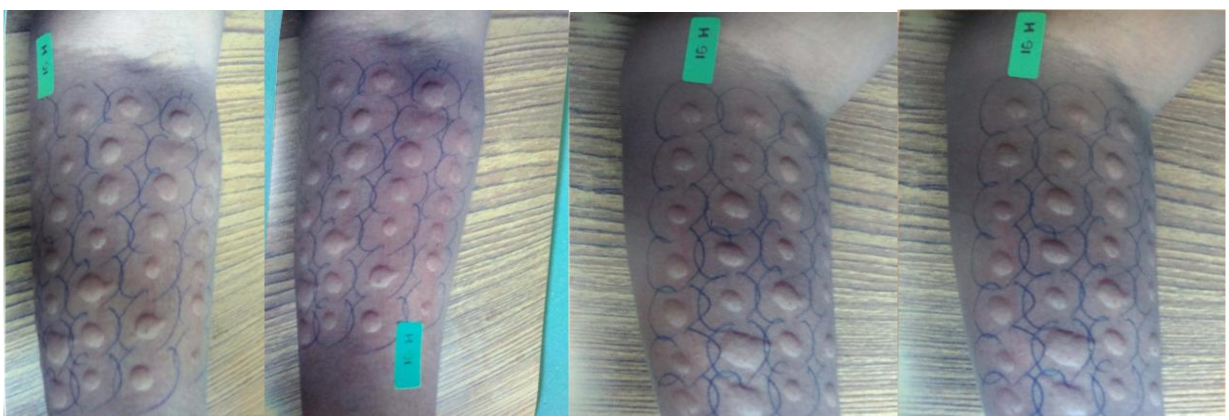
The other papers [9], [10] are about object detection from high-resolution aerial imagery using deep learning models [6], [9] and autonomous driving cars' road object detection [10]. Even though they do not specifically discuss the applications of these methods in SPT Wheal analysis, they indicate that deep learning models are versatile in detecting objects that may be adapted to identify wheals.

A multiple-headed CNN architecture is used for skin disease classification based on deep learning in the paper by Kumar et al. [11]. They have taken advantage of different domains(spectrogram, image and cepstrum) to extract more diversified knowledge about skin lesions. It shows how much diverse information about various aspects of skin conditions can be learned through DL with regards to SPT wheal analysis in future.

Chapter 3: Dataset Description

The dataset consists of two types of images, one being ‘good pics’ of the wheal test, and the other being ‘ideal pics’ of wheal test reaction. The ‘ideal pics’ depict the reaction when it tests positively against an allergen. An image consists of cropped pictures of the forearm, having multiple wheals generating more wheals to test on. The forearm has a tag indicating the arms of different patients. This tag will be used for diameter estimation of the wheals. The ‘good pics’ consists of the forearm images with reactions of SPT, that are not very prominent. It also contains image with negative SPT reactions. These images will be used for testing to identify the false positives for reading the SPT reactions.

3.1. “Good Pics” with positive and negative SPT reactions



3.2. “Ideal Pics” with positive SPT reactions of high severity

Chapter 4 - Implementation

Considering the previous methods in this field, our project puts forward a new method for SPT wheal measurement. It aims to include both 3D reconstruction methods along with Mask R-CNN, a DL model used for object detection and image segmentation. [4] This approach provides the benefits of both methods:

1. The height of the wheal can be estimated and can be presented with 3D reconstruction of wheals.
2. Mask R-CNN helps in identifying and segmenting the wheal region with precision.

The method proposed by us will follow the following procedures:

1. Developing a 3D reconstruction system for calculating wheal height.
2. Mask R-CNN model to identify, segment and measure wheal diameter.

4.1. 3D Reconstruction

The Model used to reconstruct the wheals is TripoSR. This model uses a transformer architecture to build 3D meshes from a single image. It consists of three-main components: an image encoder, an image-to-triplane decoder, and a triplane-based neural radiance field (NeRF).

The image encoder uses the transformer model to project the image into a set of vector, which contain the necessary features to build the 3D object. The latent vectors are transformed by the image-to-triplane decoder as the triplane-NeRF representation. The decoder employs self-attention and cross-attention layers to include local and global image features into the triplane representation. The NeRF model using MLPs finally predicts the color and density of 3D point in pace. The entire model is pretrained, and performs the reconstruction in less that 5 seconds. [15]

The dataset images both ‘ideal pics’ and ‘good pics’ were passed for reconstruction. The resulting model seemed to be developed better for the ideal pics when compared to the good pics. Within the ideal pics, different quantity of wheals were passed as inputs; being the whole forearm, four wheals and one wheal. The model reconstructed he forearm better than the four wheals image, and the one wheal image. The one wheal image, requires vast improvement in the reconstruction. Based on these results, it is suggested to pass the whole forearm image as input for reconstruction, as it will give as more accuracy.

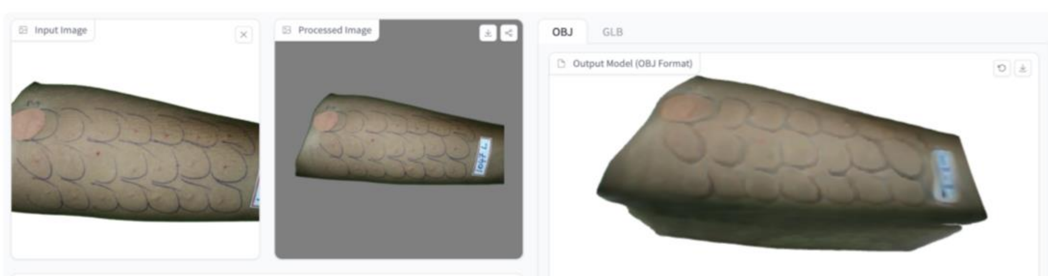


Fig 4.1. ‘good pic’ wheal reconstruction

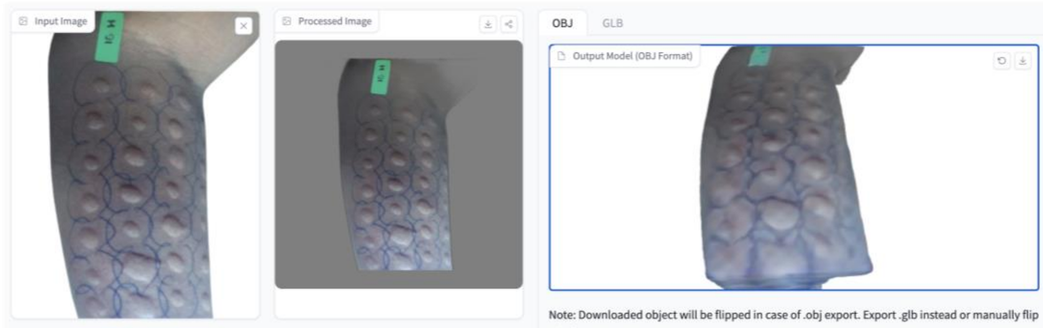


Fig 4.2. (a): Full Forearm

Fig 4.2. (b): Four Wheals

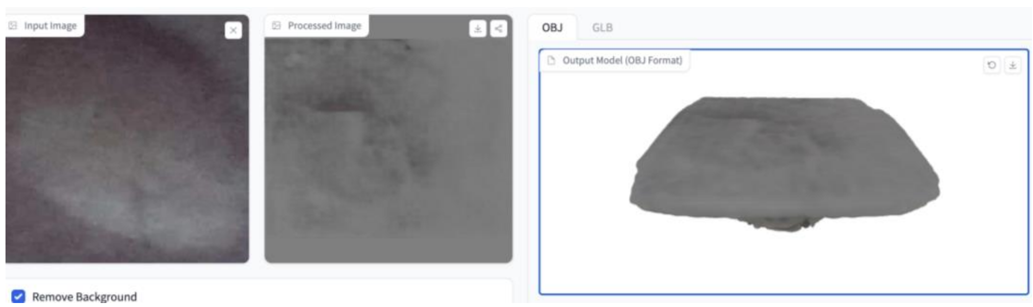


Fig 4.2. (c): One Wheal

Fig 4.2. 'ideal pic' wheal reconstruction.

4.2. Mask RCNN Model

4.2.1 Detectron2 Model

Detectron2 was devised by Facebook AI Research (FAIR) as next-generation platform for object detection and segmentation. It supports implementation of and evaluation of computer vision projects. Object Detection Algorithms such as Mask R-CNN, RetinaNet etc. can be implemented through this model.

The Model we have used for this project is Mask RCNN, which is an extension of Faster R-CNN. Faster R-CNN performs object detection on a particular image and the output is its class label and bounding box coordinates for every object in that image. The Faster RCNN utilizes a ConvNet to detect feature maps from images, whereas Mask RCNN uses ResNet 101 Architecture. The feature maps are then passed through a region proposal network (RPN), which predicts the presence of the object in that particular region. The regions are then passed through a network which predicts the class label and bounding boxes.

The prediction performed until now almost similar to the workings of Faster R-CNN, however Mask R-CNN in addition also performs segmentation. To perform the segmentation, we calculate the region of interest. The Intersection over Union (IoU) is then calculated for the regions predicted.

$$IoU = \frac{Area\ of\ the\ Intersection}{Area\ of\ the\ Union}$$

The region of interest is taken into account only if the value of IoU is above or equal to 0.5. This is performed for all regions. The mask branch is joined onto the existing architecture, which outputs a segmentation mask of size 28 x 28 for each region. This region can be scaled up for making inferences.

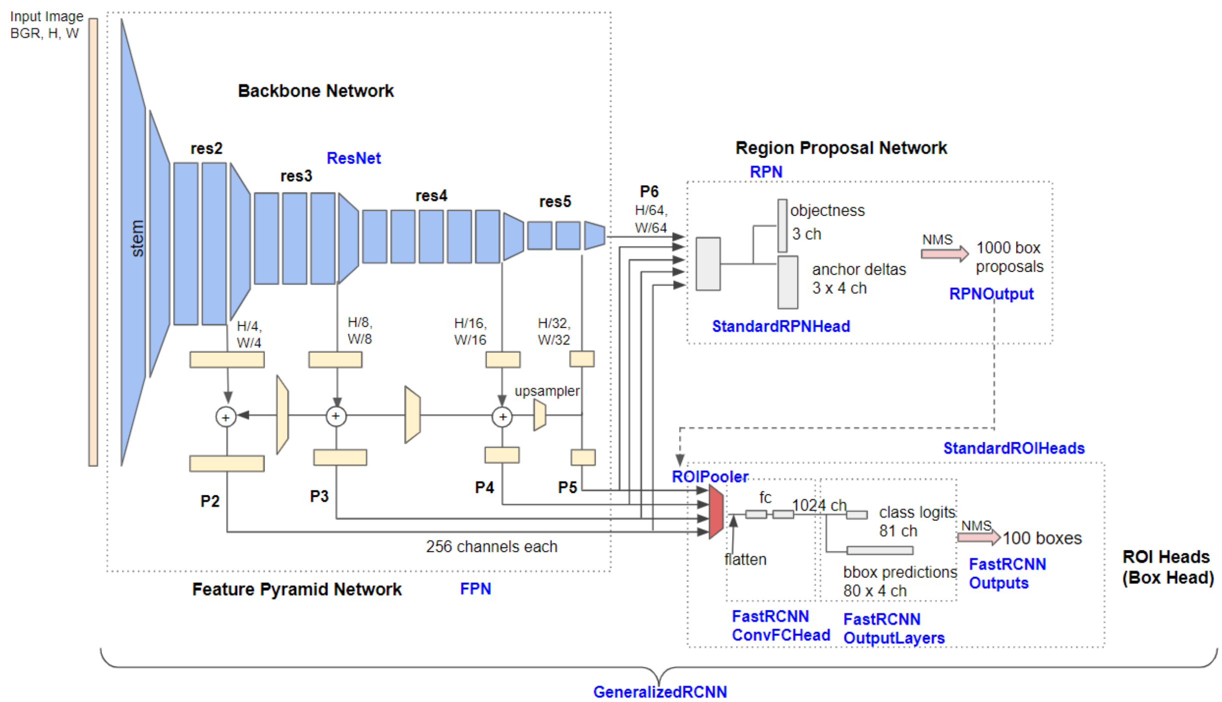


Fig 4.3. Dectron2 Architecture

4.2.2 Roboflow

Our model uses the detectron2 model trained on the ‘ideal pics’, which was manually annotated using Roboflow, a framework for creating computer vision models. The framework allows us to download the annotations in any required format which is the COCO segmentation format for out model. Since the number of images used for the model was less, we have also augmented the image by changing the saturation of the images (using Roboflow), giving us a total of 11 images with the split: 9 test images, 1 valid image, 1 train image.

There were two class in the annotated image the first one being the ‘flares’ region, and the second being the ‘mark region’. The ‘flares’ indicated the wheals on the hand due to the allergic reaction, while the ‘mark’ indicates a sticker placed on the patient’s hand. The ‘mark’ region would later be used for measuring the diameter of the largest flare, which helps in the analysis of the SPT to be performed.

Fig 4.4. (a) Train

Fig 4.4. (b) Valid

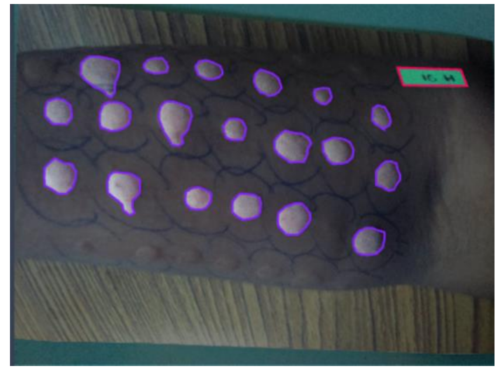


Fig 4.4. (c) Test

Fig 4.4. Annotated Images

The annotated COCO segmentation file, has been imported directly onto the training file using the 'roboflow' module in python. This module helps us access a particular model, dataset etc. using the API Key provided by the Roboflow Account. Through this method the dataset can be accessed in any system and does not require local installments.

4.2.3 Hardware

To make it easier to train our model, and for stronger GPU the IDE used for this project is Google Colab. The Hardware pertaining to the project is NVIDIA's Tesla T4 GPU with 16GB RAM connected by a 256-bit memory interface.

```
Wed May 22 13:51:46 2024
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| NVIDIA-SMI 535.104.05             Driver Version: 535.104.05   CUDA Version: 12.2     |
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Fig 4.5. NVIDIA Features

4.2.4 Training

To check if the detectron2 model worked properly, we tested the model on a sample image. The model delivered accurate results with masks and bounding boxes annotated on the image.

Fig 4.6. (a) Before

Fig 4.6. (b) After

Fig 4.6. Testing on pretrained model

The model was then retrained on our dataset with the parameters:

- ARCHITECTURE: mask_rcnn_R_101_FPN_3x
- CONFIG_FILE_PATH: COCO-InstanceSegmentation/mask_rcnn_R_101_FPN_3x.yaml
- MAX_ITER: 2000
- EVAL_PERIOD: 200
- BASE_LR: 0.001
- NUM_CLASSES: 3 (including the background as a class)
- MODEL_WEIGHTS: model_zoo.get_checkpoint_url("COCO-InstanceSegmentation/mask_rcnn_R_101_FPN_3x.yaml")
- ROI_HEADS.BATCH_SIZE_PER_IMAGE: 64
- EVAL_PERIOD: 200
- NUM_WORKER: 2
- IMGS_PER_BATCH: 2
- MASK_FORMAT: bitmask

The result of retraining the model can be seen here below:

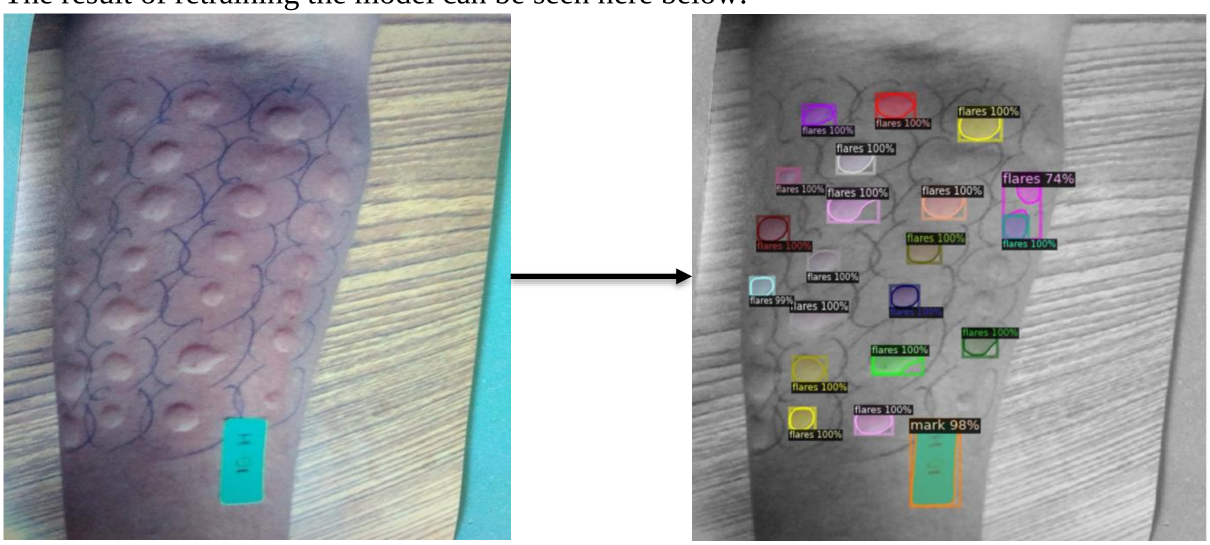

Fig 4.7. (a) Before

Fig 4.7. (b) After

Fig 4.7. Retraining the model on new dataset

4.3. Diameter Calculations

For the diameter calculations of the wheal, the area estimation on the ‘mark’ class is used. We have extracted the pixels occupied by each masks, along with their classes to determine the area occupied by each flare. The area occupied by the mask is estimated to be 0.75cm, with which the flare areas have been approximated.

Using the formula:

$$Area\ of\ circle = \pi r^2$$

The radius of each flare region was estimated along with it is diameter. For any SPT test, only the flare with the largest diameter is taken into account, which was estimated to be 0.8336 cm.

Chapter 5 – Proposed Methodology

Building upon the implementation we propose an integration of both the 3D Reconstruction model as well as the MASK RCNN model. The integration of them through an app using live-time images makes it extremely beneficial for further clinical use.

5.1. Integration

The integration of the models is seemly for easy usage for clinical purposes. Though, the hardware of both models have to be aligned to increase performance of the system as singular model. We propose the integration through the medium of an app which is deployed and can be used in real time.

The app should firstly provide a medium to capture image in real-time. These images' should be registered before being used in the newly trained model, which can be done in the back-end. The captured image can then be reconstructed using the TripoSR model, and displayed to the user for them to make suitable predictions. Along with the reconstruction, the wheal with the largest diameter should be indicated.

With both the reconstruction, and estimation of the largest diameter the model will be completely suitable to use it for clinical purposes.

Chapter 6 – Results and Discussion

The model's performance is evaluated using Average Precision (AP) and Average Recall (AR).

Average Precision (AP) is a measure of the precision and recall trade-off for object detection and segmentation. It's calculated for different Intersection over Union (IoU) thresholds. Average Recall (AR) measures the ability of the model to find all relevant instances in the dataset. Higher recall indicates fewer missed detections.

There two measures evaluated including thee 'Bounding Box' and the 'Segmentation'. The bounding box metrics evaluates the models ability to locate the region of interests and annotate rectangular boxes around it. While, Segmentation metrics refers to the model's ability to recognize the exact shape of the wheal and form pixel-wise masks.

Bounding Box Metrics for Each Class

Class	AP @ IoU=0.50:0.95	AP @ IoU=0.50	AP @ IoU=0.75	AR @ IoU=0.50:0.95	AR @ IoU=0.50	AR @ IoU=0.75
flares	31.401	95.316	50.248	0.311	0.953	0.502
mark	60.000	95.316	50.248	0.487	0.953	0.502

Segmentation Metrics for Each Class

Class	AP @ IoU=0.50:0.95	AP @ IoU=0.50	AP @ IoU=0.75	AR @ IoU=0.50:0.95	AR @ IoU=0.50	AR @ IoU=0.75
flares	37.118	95.316	54.512	0.363	0.953	0.545
mark	70.000	95.316	54.512	0.563	0.953	0.545

Fig 6.1. Summary Tables for Each Metric Type

Both the 'bounding box' and 'segmentation' metrics show the highest precision at IoU=0.50, indicating that accurate detections can be made on objects loose bounding boxes. Precision decreases at higher threshold, which is normal due to stricter overlapping criteria As seen from the metrics, the model can easily detect the 'mark' as compared to the 'flare' with the highest precision of both at IoU=0.50 being 95.5316.

The model seems to perform well in detecting and segmenting objects. It can be improved further by providing large number of images to work with to get more accurate results.

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